

Introduction

Baroid Core Homework 18

Assessment Feedback

Congratulations, you passed.

Total score: 49 out of 58, 84%

Question Feedback

Question

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Select the correct definition of formation damage: _____

- ☐ Any process that increases the natural porosity of a formation/rock
- ☒ Any process that reduces the natural permeability of a formation/rock
- ☐ Any process that maintains the natural permeability of a formation/rock
- ☐ Any process that changes the natural structure of a formation/rock

1 out of 1

Question

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Formation damage can be caused by two types of mechanisms, they are physical and chemical mechanisms.

2 out of 2

Question

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The reservoir is the rock that contains the hydrocarbons.

1 out of 1

Question

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Formation fluids are specially engineered fluids that are used to drill through the payzone.

- ☒ True
☐ False

0 out of 1

Question

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The occurrence of formation damage will lower the production rate of a reservoir.

- ☒ True
☐ False

1 out of 1

Question

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Porosity is the volume of the rock which can be filled by formation fluids.

1 out of 1

Question

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The larger the porosity of a formation, the more resistant it will be to formation damage.

- ☐ True
☒ False

0 out of 1

Question

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Select the main causes of limited production rate

- ☒ Formation damage
- ☐ Low formation salinity
- ☒ Low formation permeability
- ☒ High oil viscosity
- ☒ Low formation pressure
- ☐ High oil API gravity
- ☐ Low formation temperature

1 out of 1

Question

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Radial flow of the produced fluids (into the wellbore) creates an increase ▼ of the flow velocity near the wellbore

1 out of 1

Question

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Select from the following options, all the relevant formation damage mechanisms

- ☐ Clay inhibition
- ☒ Emulsion blockage
- ☒ Polymer adhesion
- ☐ Fluid density
- ☒ Particle size distribution (PSD)
- ☒ Clay migration and dispersion
- ☒ Particle plugging

0 out of 4

Question

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From the following list, select all the different particles than can cause damage inside the reservoir:

- ☒ Polymers not soluble in acid
- ☒ Barite
- ☐ Calcium carbonate
- ☒ Perforation debris
- ☐ Polymers soluble in acid
- ☐ All of the above

3 out of 3

Question

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Regarding clays, list the clays discussed in class and what is the behavior expected for each one of them and how to prevent it

Smectite Takes water and swell, prevent with potassium ▼

migration and dispersion

Kaolinite	Prone to swelling if potassium is present, avoid using >3% v/v of potassium	▼
Illite	Prone to migration and dispersion, avoid dispersants and stabilize with potassium	▼

2 out of 3

Question

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The way polymers are mixed in the fluid system is not important since they cannot generate any formation damage because they are very small.

- ☐ True
☒ False

1 out of 1

Question

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The filtrate from drilling fluids needs to be compatible with the water from the formation, to avoid the formation of precipitates.

- ☒ True
☐ False

1 out of 1

Question

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The excessive use of emulsifiers ▼ can lead to the formation of crude-brine emulsions that ultimately will translate into formation damage

1 out of 1

Question

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One of the ways to protect the reservoir from damage is the buildup of a thick filter cake that can increase the invasion around the wellbore

- ☐ True
☒ False

1 out of 1

Question

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Which is probably more damaging to a payzone?

- ☐ A dirty mud with 20% solids
- ☒ A dirty brine with 10% solids

1 out of 1

Question

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Reservoir fluids can be defined as Fluids we expose to the reservoir, that help prepare, repair, cleanout and complete the wellbore

- ☒ True
- ☐ False

1 out of 1

Question

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Which of the following fluids can be used in reservoir while drilling and completing (select all applicable answers)?

- ☒ Workover fluids
- ☒ Completion brines
- ☒ Reservoir drilling fluids
- ☐ Payzone hydrocarbons
- ☐ Unfiltered sea water
- ☒ Payzone water

3 out of 3

Question

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Which products might you want to use in reservoir fluids, for as long as they're compatible with the payzone?

- ☒ Bactericides
- ☐ Clays
- ☒ Pipe dopes ☒
- ☒ Lubricants
- ☐ Sticky polymers (PAC)
- ☒ Viscosifiers

3 out of 3

Question

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Completion fluids are always clear brines with no viscosity

- ☐ True
- ☒ False

1 out of 1

Question

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The goal of a displacement is to remove all the products/by products of the drilling process before production

- ☒ True
- ☐ False

1 out of 1

Question

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Reservoir fluids are specifically designed to:

- ☒ Minimize formation damage
- ☐ Increase the pH of the formation
- ☒ Maintain well producibility
- ☐ All of the above

2 out of 2

Question

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Completions and workovers require clean fluids for maximum effectiveness, and that is the goal of the filtration services

- ☒ True
- ☐ False

1 out of 1

Question

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A filtration unit can help to recover clean potable water (free of solids and chemicals) from completion and workover fluids

- ☐ True
- ☒ False

1 out of 1

Question

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Which of the following are characteristics of a reservoir drill-in fluid?

- ☒ Efficiently transports cuttings to the surface
- ☒ Minimizes potential formation damage
- ☒ Maintains hydrostatic overbalance to control the well
- ☐ Minimizes ROP and increases costs
- ☐ Same characteristics as any top hole drilling fluid

3 out of 3

Question

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When using a Dril-N fluid system, we know we have preserved the reservoir's producibility if the near wellbore rock flows the same after the well is completed as it did prior to the drilling/completion

- ☒ True
- ☐ False

1 out of 1

Question

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What are characteristics of a poorly-designed reservoir fluid?

- ☐ Controls bottom hole pressures
- ☐ Optimizes cuttings carrying capacity
- ☒ Allows large quantities of foreign material to pass into the payzone
- ☒ Is not tested and potential damage prior use is unknown
- ☒ Maximizes ROP by employing high filtrate losses

Question

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Does a fracked completion normally require a specially engineered drill-in fluid?

- ☐ Yes
- ☒ No

1 out of 1

Question

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Does an open hole completion normally require a specially-engineered drill-in fluid?

- ☒ Yes
- ☐ No

1 out of 1

Question

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Which of the following are potentially damaging to the payzone?

- ☒ Unfiltered Seawater
- ☐ All of the above
- ☐ Filtered brine
- ☒ Drill solids
- ☒ Mud filtrate
- ☒ Mud products

3 out of 4

Question

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We prefer to use ground marble for drill-in fluid bridging because

- ☐ It is less costly than ground limestone
- ☐ It looks clean and white
- ☒ It is harder than limestone
- ☐ All of the above

1 out of 1

Question

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Kozeny's Equation estimates a reservoir's pore throat diameter in microns when the rock permeability is known.

- ☒ True
- ☐ False

1 out of 1

Question

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On a particle size distribution, what is the meaning of the D_{50} value of a bridging material analysis?

- ☐ 50% of the particles are larger than the D_{90} value
- ☒ 50% of the particles are smaller than the D_{50} value
- ☐ 50% of the particles are pure marble
- ☐ None of the above

1 out of 1

Question

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What concentration of bridging material has been found to be most effective for drill-in fluids?

- ☐ 20% by volume
- ☒ Between 20 and 30 lb/bbl
- ☒ Between 30 and 45 lb/bbl
- ☐ Between 35 and 50 lb/bbl

0 out of 1

Question

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Which return permeability value would be considered to be good by most operators?

- ☐ 70%
- ☐ 80%
- ☐ 85%

☒ >90%

1 out of 1

Question

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High Performance Invert Emulsion Fluids can make excellent drill-in fluids, given that the appropriate solids content is achieved

☒ True

☐ False

1 out of 1

Question

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Any type of LCM can be used if there are losses in the payzone

☐ True

☒ False

1 out of 1

TQ